

A Constant Finite-Center LCQFT Extension and the Accountability Premise

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Abstract

Let $\mathcal{A}$ be a BFV locally covariant quantum field theory valued in unital $*$ -algebras, and let $Z_q = \mathbb{C}^q$. The constant finite-center extension

$$\mathcal{A}_q(M) = \mathcal{A}(M) \otimes Z_q, \quad \mathcal{A}_q(\psi) = \mathcal{A}(\psi) \otimes \text{id}_{Z_q}$$

preserves functoriality, injective morphisms, and the time-slice axiom. Its relative Cauchy evolution is

$$\text{rce}_q[h] = \text{rce}[h] \otimes \text{id}_{Z_q}.$$

If dynamical locality is understood in the Fewster-Verch RCE sense, and the target tensor product identifies $\mathcal{A}(M) \otimes \mathbb{C}^q$ with $\bigoplus_{j=1}^q \mathcal{A}(M)$, then dynamical locality is inherited from $\mathcal{A}$: both the kinematic and dynamical local subobjects are tensored by Z_q . Nevertheless any record that factors through a character $\pi_j : Z_q \rightarrow \mathbb{C}$ is blind to the nonselected finite-center directions. Thus BFV functoriality, time-slice, RCE, and RCE dynamical locality alone do not reconstruct hidden finite central rank. A positive result must add the exact premise that removes or measures the center: extended locality, local definiteness, center reduction, indecomposability, or a center-separating measurement channel.

Status. This is non-frozen external-review metadata for Team 1. It is intended as a small claim that an LCQFT referee can classify as known, false, missing-hypothesis, or useful. It is not a new construction of LCQFT, and it does not modify any frozen Team 1 manuscript, Lean archive, public mirror manifest, or SHA-256 package.

1 Setup

Let **Loc** denote the usual BFV category of globally hyperbolic spacetimes and admissible embeddings, and let **Alg** be a category of unital $*$ -algebras with injective morphisms. A locally covariant theory is a functor

$$\mathcal{A} : \text{Loc} \longrightarrow \text{Alg}.$$

It obeys the time-slice axiom if $\mathcal{A}(\psi)$ is an isomorphism whenever ψ is a Cauchy morphism. Relative Cauchy evolution is the automorphism of $\mathcal{A}(M)$ obtained from the time-slice zig-zag associated with a compactly supported metric perturbation.

Fix $q \geq 2$, let $Z_q = \mathbb{C}^q$, and define

$$\mathcal{A}_q(M) = \mathcal{A}(M) \otimes Z_q, \quad \mathcal{A}_q(\psi) = \mathcal{A}(\psi) \otimes \text{id}_{Z_q}.$$

For a character $\pi_j : Z_q \rightarrow \mathbb{C}$, define the quotient-visible natural map

$$Q_{j,M} = \text{id}_{\mathcal{A}(M)} \otimes \pi_j : \mathcal{A}_q(M) \rightarrow \mathcal{A}(M).$$

Theorem 1 (constant finite-center extension). *If $\mathcal{A}$ is a BFV locally covariant theory with injective morphisms, then $\mathcal{A}_q$ is also a BFV locally covariant theory with injective morphisms. If $\mathcal{A}$ obeys the time-slice axiom, then so does $\mathcal{A}_q$, and*

$$\text{rce}_q[h] = \text{rce}[h] \otimes \text{id}_{Z_q}.$$

Any operational, sector, recovery, modular, or RCE record factoring through $Q_{j,M}$ is insensitive to the nonselected central idempotents and to every nonconstant central contrast in $\ker \pi_j$.

Proof. Functoriality, identities, and composition are componentwise. Injectivity is preserved because tensoring with the finite-dimensional algebra Z_q is exact here; equivalently, $\mathcal{A}(M) \otimes \mathbb{C}^q \simeq \bigoplus_{j=1}^q \mathcal{A}(M)$, and $\mathcal{A}(\psi) \otimes \text{id}_{Z_q}$ is the direct sum of q copies of $\mathcal{A}(\psi)$. If ψ is Cauchy, $\mathcal{A}(\psi)$ is an isomorphism, so its finite direct sum is an isomorphism. Thus the time-slice axiom is preserved. The relative Cauchy evolution in $\mathcal{A}_q$ is obtained by applying the same time-slice zig-zag with every arrow tensored by id_{Z_q} , giving the stated formula.

Finally, if $R_q = R \circ Q_{j,M}$, then for $a \in \mathcal{A}(M)$ and $z, z' \in Z_q$ with $\pi_j(z) = \pi_j(z')$,

$$R_q(a \otimes z) = R(a \pi_j(z)) = R(a \pi_j(z')) = R_q(a \otimes z').$$

Thus the record is constant on the hidden finite-center fiber. $\square$

2 Dynamical Locality

Fewster-Verch dynamical locality compares the kinematic local subobject

$$\mathcal{A}_{\text{kin}}(M; O) = \mathcal{A}(\iota_O)(\mathcal{A}(M|_O))$$

with the dynamical subobject

$$\mathcal{A}_{\text{dyn}}(M; O) = \{a \in \mathcal{A}(M) : \text{rce}_M[h](a) = a \text{ for all } h \text{ supported in } O^\perp\}.$$

The theory is dynamically local when these two subobjects agree for all admissible O .

Proposition 1 (inheritance of RCE dynamical locality). *Assume $\mathcal{A}$ is dynamically local in the Fewster-Verch RCE sense, and assume the tensor product in the target algebra category identifies $\mathcal{A}(M) \otimes \mathbb{C}^q$ with $\bigoplus_{j=1}^q \mathcal{A}(M)$. Then the constant finite-center extension $\mathcal{A}_q$ is dynamically local in the same sense.*

Proof. For every admissible $O \subset M$,

$$(\mathcal{A}_q)_{\text{kin}}(M; O) = \mathcal{A}_{\text{kin}}(M; O) \otimes \mathbb{C}^q$$

by functoriality of the inclusion $\iota_O : M|_O \rightarrow M$. By Theorem 1,

$$\text{rce}_q[h] = \text{rce}[h] \otimes \text{id}_{\mathbb{C}^q}.$$

Under the finite direct-sum identification, an element of $\mathcal{A}_q(M)$ is a tuple $(a_1, \dots, a_q)$. It is fixed by all $\text{rce}_q[h]$ with $\text{supp } h \subset O^\perp$ exactly when every a_j is fixed by the corresponding $\text{rce}[h]$. Hence

$$(\mathcal{A}_q)_{\text{dyn}}(M; O) = \mathcal{A}_{\text{dyn}}(M; O) \otimes \mathbb{C}^q.$$

If $\mathcal{A}_{\text{kin}}(M; O) = \mathcal{A}_{\text{dyn}}(M; O)$, then $(\mathcal{A}_q)_{\text{kin}}(M; O) = (\mathcal{A}_q)_{\text{dyn}}(M; O)$. $\square$

Proposition 2 (extended locality and local definiteness are positive exits). *Let $q > 1$. Suppose a spacetime M contains two nonempty causally disjoint admissible regions O_1, O_2 , and let their kinematic images in $\mathcal{A}_q(M)$ be denoted by $(\mathcal{A}_q)_{\text{kin}}(M; O_i)$. Then*

$$\mathbb{C}1_{\mathcal{A}(M)} \otimes Z_q \subset (\mathcal{A}_q)_{\text{kin}}(M; O_1) \cap (\mathcal{A}_q)_{\text{kin}}(M; O_2).$$

Consequently $\mathcal{A}_q$ violates any extended-locality axiom requiring causally disjoint nonempty local algebras to intersect only in scalars. Likewise, if a local-definiteness axiom requires the intersection of algebras for regions shrinking to a point to be only $\mathbb{C}1$, then $\mathcal{A}_q$ violates that axiom for the same reason: the constant finite center remains present in every local kinematic algebra.

Proof. For every admissible inclusion $\iota_O : M|_O \rightarrow M$, the element $1_{\mathcal{A}(M|_O)} \otimes z \in \mathcal{A}(M|_O) \otimes Z_q$ is mapped by $\mathcal{A}(\iota_O) \otimes \text{id}_{Z_q}$ to $1_{\mathcal{A}(M)} \otimes z$. Hence the same central copy $\mathbb{C}1_{\mathcal{A}(M)} \otimes Z_q$ lies in the kinematic image of every nonempty region. If $q > 1$, this copy contains non-scalar central idempotents, so the intersection of two causally disjoint local algebras is not just $\mathbb{C}1$. The shrinking-region statement is identical: every algebra in the shrinking family contains the same central copy. $\square$

3 Positive Exits

Proposition 1 is narrow. It says that the RCE-defined equality of kinematic and dynamical local subobjects does not by itself remove a constant finite center. Proposition 2 gives the matching positive exit: stronger physical assumptions can remove it.

Exit premise	Effect on $\mathcal{A}_q$ for $q > 1$
Extended locality	If algebras of causally disjoint nonempty regions may intersect only in scalars, $\mathcal{A}_q$ fails because every local algebra contains the common central copy of Z_q .
Local definiteness	If the intersection of local algebras shrinking to a point must be $\mathbb{C}$, $\mathcal{A}_q$ fails because Z_q remains in every local algebra.
Center reduction	If the framework quotients or reduces away central degrees of freedom, as in reduced Maxwell-type constructions, the missing physical-exclusion premise has been supplied.
Center-separating measurement	If the probe/readout is tomographically complete on Z_q , the data do not factor through a single character π_j ; the quotient-visible hypothesis is false.

4 Referee Question

The classification question is precise:

Is the constant finite-center extension above already excluded by the standard LCQFT hypotheses intended in the target setting? If so, name the axiom or theorem. If not, BFV functoriality, time-slice, RCE, and RCE dynamical locality alone do not supply finite-center accountability.

The desired external response is not encouragement. It is one of: known theorem, false statement, missing positive hypothesis, physically too broad, or useful expository no-go.

References

- [1] R. Brunetti, K. Fredenhagen, and R. Verch, The generally covariant locality principle: a new paradigm for local quantum field theory, *Communications in Mathematical Physics* **237** (2003), 31–68; [arXiv:math-ph/0112041](#).
- [2] C. J. Fewster and R. Verch, Dynamical locality and covariance: what makes a physical theory the same in all spacetimes?, *Annales Henri Poincaré* **13** (2012), 1613–1674; [arXiv:1106.4785](#).
- [3] C. J. Fewster, The paradigm of local covariance, in *Quantum Field Theory and Gravity*, Birkhauser, 2012; see also [arXiv:1105.6202](#).

- [4] C. J. Fewster and R. Verch, Quantum fields and local measurements, *Communications in Mathematical Physics* **378** (2020), 851–889; [arXiv:1810.06512](#).
- [5] C. J. Fewster and R. Verch, Measurement in quantum field theory, *Lecture Notes in Physics* **1017** (2024), 1–45; [arXiv:2304.13356](#).
- [6] C. J. Fewster and B. Lang, Dynamical locality of the free Maxwell field, *Annales Henri Poincare* **17** (2016), 401–436; [arXiv:1403.7083](#).